The Semantic Field Coherence Score (SFCS): An Observability Metric for Early Detection of Semantic Field Degradation and Domain Drift in AI Systems

Abstract

Operational AI assurance requires mechanisms capable of detecting semantic instability before it manifests as observable failures. Existing monitoring approaches primarily focus on accuracy, safety, factuality, benchmark performance, and hallucination detection, providing limited visibility into whether an AI system remains anchored within its intended semantic field.

This paper introduces the **Semantic Field Coherence Score (SFCS)**, an observability metric designed to quantify semantic field coherence across repeated AI outputs. Rather than measuring truth, correctness, or logical agreement, SFCS measures the degree to which outputs remain anchored within a shared semantic field.

To evaluate the metric, we developed the **SFCS Validation Workbench** and conducted controlled divergence experiments, contradiction analysis, domain-shift testing, and cross-domain drift validation spanning finance, artificial intelligence, medical, and legal domains.

Experimental results demonstrated systematic degradation of SFCS under progressive semantic divergence and domain drift conditions while contradictory statements within the same semantic field maintained high coherence scores. Cross-domain experiments revealed comparable degradation trajectories across multiple domains.

These findings suggest that semantic field degradation may be observable before complete semantic collapse occurs, positioning SFCS as a potential early-warning signal for emerging domain drift and instability in operational AI systems.

Keywords: AI Observability, Operational AI Assurance, Semantic Field Coherence, Domain Drift Detection, Semantic Stability, Risk Monitoring, AI Auditing

1. Introduction

As AI systems become increasingly integrated into operational, scientific, and decision-support environments, monitoring requirements extend beyond traditional performance metrics.

Current AI observability approaches primarily focus on:

- accuracy
- safety
- hallucination detection
- benchmark performance
- reliability

While valuable, these approaches provide limited visibility into whether an AI system remains within its intended conceptual domain during operation.

A system may continue producing syntactically plausible outputs while gradually drifting away from its intended semantic field. Such transitions may occur long before visible failures emerge and may therefore remain undetected by conventional monitoring approaches.

This creates an observability gap between normal operation and observable failure.

The central question addressed in this paper is:

Can semantic field coherence be operationalized as a measurable observable?

To answer this question, we introduce the **Semantic Field Coherence Score (SFCS)**, a metric designed to quantify the degree to which multiple outputs remain anchored within a shared semantic field.

The guiding principle of this work is straightforward:

Stability cannot be controlled if it cannot first be observed.

1.1 Related Work

AI monitoring and observability have traditionally focused on performance metrics, safety evaluation, factual correctness, hallucination detection, and dataset shift analysis. While these approaches provide valuable operational signals, they offer limited visibility into whether a system remains anchored within its intended semantic field.

Research on semantic change has demonstrated that meaning representations evolve and shift within embedding spaces over time (Hamilton et al., 2016; Schlechtweg et al., 2020). These studies establish that semantic relationships can be quantified within vector-space representations and provide an important foundation for measuring semantic behavior computationally.

Within machine learning, substantial work has focused on dataset shift and distribution shift detection. Gretton et al. (2012) introduced statistical approaches for comparing distributions, while Rabanser et al. (2019) demonstrated the importance of detecting shifts before model performance deteriorates. However, these approaches primarily operate at the dataset or distribution level rather than at the level of semantic field coherence within generated outputs.

Recent work on embedding representations has further highlighted the complexity of semantic behavior in vector spaces. Ethayarajh (2019) showed that contextualized embeddings exhibit substantial variation across contexts, emphasizing the importance of understanding representation-level effects when designing observability metrics.

The present work differs from these approaches by focusing on semantic field coherence as an operational observability signal. Rather than measuring dataset shift, lexical change, truthfulness, or logical agreement, SFCS is designed to quantify the degree to which outputs remain anchored within a shared semantic field. To our knowledge, semantic field coherence itself has rarely been treated as an explicit observability target for operational AI assurance.

Contributions

This paper makes five primary contributions:

C1. Introduction of the Semantic Field Coherence Score (SFCS).

We introduce SFCS as an observability metric designed to quantify the degree to which AI outputs remain anchored within a shared semantic field. Unlike metrics focused on truth, factuality, or logical agreement, SFCS measures semantic field coherence.

C2. Development of the SFCS Validation Workbench.

We present a dedicated experimental environment for systematically evaluating semantic field coherence, semantic degradation, and domain drift behavior under controlled conditions.

C3. Empirical validation of semantic field sensitivity.

Through controlled divergence experiments, we demonstrate that SFCS exhibits systematic degradation as semantic divergence increases.

C4. Characterization of semantic coherence under contradiction and domain shift.

Experimental results show that contradictory statements may remain highly coherent when anchored within the same semantic field, while domain-shift scenarios produce substantially lower coherence values. This distinction suggests that SFCS is sensitive to semantic field structure rather than logical agreement alone.

C5. Cross-domain validation of semantic degradation dynamics.

Using drift trajectories in finance, artificial intelligence, medical, and legal domains, we demonstrate comparable degradation patterns across multiple semantic fields, suggesting that SFCS captures a domain-independent observability signal related to semantic field coherence.

2. The Semantic Drift Observability Gap

2.1 Monitoring Versus Failure Detection

Most current AI monitoring approaches detect failures after they emerge.

Examples include:

- hallucination detection
- factuality checking
- safety violation monitoring
- benchmark evaluation

These approaches focus on outcomes rather than the gradual semantic processes that may precede failure.

2.2 Semantic Field Degradation

We define **semantic field degradation** as the progressive loss of coherence within an active semantic field.

Rather than occurring as a binary failure event, degradation may emerge gradually through:

- conceptual drift
- semantic fragmentation
- weakening field coherence
- transition into unrelated semantic regions

2.3 Domain Drift as an Observability Problem

Domain drift occurs when outputs progressively move away from the intended conceptual region.

Examples include:

Plain Text
Cardiology
↓
Medicine
↓
Biology
↓
Physics
↓
History

or

Plain Text
Machine Learning
↓
Statistics
↓
Mathematics
↓
Physics
↓
Random Topics

The challenge is not merely detecting failure but observing the trajectory before failure becomes visible.

2.4 Research Questions

RQ1

Can semantic field coherence be operationalized as a measurable observable?

RQ2

Does SFCS respond systematically to increasing semantic divergence?

RQ3

Does SFCS measure semantic field coherence rather than logical agreement?

RQ4

Is SFCS sensitive to semantic field fragmentation and domain drift?

RQ5

Can SFCS function as a practical observability signal for operational AI assurance?

RQ6

Can semantic field degradation be observed before complete semantic collapse occurs?

3. Semantic Field Stability

3.1 Semantic Fields

A semantic field is defined as a coherent region of meaning-space characterized by related concepts, relationships, and contextual structures.

Example:

Plain Text
Machine Learning
Model Training
Generalization
Feature Engineering

Overfitting

constitutes a coherent semantic field.

Outputs originating from a common semantic field are expected to exhibit greater semantic convergence than outputs originating from unrelated conceptual domains.

3.2 Semantic Field Coherence

Semantic field coherence refers to the degree to which multiple outputs remain anchored within a shared conceptual region.

High coherence indicates semantic convergence.

Low coherence indicates fragmentation or semantic drift.

3.3 Hypotheses

H1

Increasing semantic divergence results in decreasing semantic field coherence.

H2

Outputs sharing a common semantic field produce higher SFCS values.

H3

Domain shift and semantic fragmentation reduce SFCS values.

4. The Semantic Field Coherence Score

4.1 Formal Definition

Given a set of outputs:

Plain Text

$X_1, X_2, \ldots, X_n$

Each output is transformed into a semantic embedding:

Plain Text
E(X)

Pairwise cosine similarity is calculated as:

Plain Text
$sim(i,j) = \cos(E(X_i), E(X_j))$

The Semantic Field Coherence Score is defined as:

Plain Text
SFCS = Mean Pairwise Similarity

across all unique output pairs.

4.2 Measurement Chain

Plain Text
Repeated Executions
↓
Multiple Outputs
↓
Sentence Embeddings
↓
Pairwise Similarities
↓
SFCS
↓
Observed Semantic Field Coherence

4.3 Interpretation

SFCS measures:

The degree to which outputs remain anchored within a shared semantic field.

SFCS does not attempt to measure:

- truth
- factual correctness
- logical validity
- reasoning-path equivalence

5. SFCS Validation Workbench and Experimental Artifacts

To evaluate the observable properties of the Semantic Field Coherence Score (SFCS), a dedicated experimental environment was developed. Rather than evaluating the metric through isolated examples, the objective was to create a reproducible framework for systematically characterizing semantic field coherence, semantic degradation, and domain drift behavior.

The resulting implementation, referred to as the **SFCS Validation Workbench**, provides an extensible environment for controlled experimentation and metric evaluation.

5.1 Implementation

The SFCS Validation Workbench was implemented in Python using sentence-transformer embeddings and pairwise cosine similarity calculations.

The implementation consists of multiple components that support metric evaluation, drift simulation, result analysis, and figure generation.

The experimental framework includes:

- SFCS Validation Workbench
- Controlled Divergence Validation Suite
- Property Exploration Validation Suite
- Domain Drift Evaluation Suites
- Cross-Domain Validation Dashboard
- Result Export Utilities
- Figure Generation Tools

All experiments reported in this paper were executed using the same implementation framework to ensure methodological consistency.

5.2 Measurement Workflow

The validation process follows a consistent measurement pipeline:

Plain Text
AI Outputs
↓
Sentence Embeddings
↓
Pairwise Similarity Matrix
↓
SFCS Computation
↓
Coherence Analysis
↓
Validation Results

For each experiment, multiple outputs are transformed into embedding vectors. Pairwise cosine similarity is then calculated between all outputs, and the resulting similarities are aggregated into a single Semantic Field Coherence Score.

5.3 Validation Workbench

The Validation Workbench was designed to evaluate the responsiveness of SFCS under controlled semantic conditions.

The framework supports:

- highly coherent semantic fields
- progressively degraded semantic fields
- contradictory statements
- synonym variation
- domain-shift scenarios
- cross-domain drift trajectories

This approach enables the controlled examination of metric behavior before deployment in more complex operational environments.

5.4 Domain Drift Evaluation Suites

To investigate domain-independent behavior, dedicated drift-evaluation suites were developed across multiple semantic domains.

The current implementation includes:

- Finance Drift Suite
- Artificial Intelligence Drift Suite
- Medical Drift Suite
- Legal Drift Suite

Each suite models progressive semantic transitions from an internally coherent reference domain toward increasingly distant conceptual regions.

The objective is not to measure task performance but to observe how semantic field coherence changes as conceptual drift accumulates.

5.5 Experimental Data and Reproducibility

All experiments produce structured outputs suitable for independent verification and analysis.

Generated artifacts include:

- validation datasets
- domain-drift datasets
- cross-domain evaluation results
- exported CSV result files
- publication figures
- figure-generation scripts

These artifacts allow the reported experiments to be reproduced and independently evaluated.

The implementation and experimental artifacts are intended to support future work in AI observability, semantic drift monitoring, and operational assurance workflows. All datasets and figures reported in this study were generated using the software artifacts described in this section.
github

6. Experimental Design

6.1 Controlled Divergence Validation

Four progressively divergent semantic scenarios were evaluated:

- Highly Stable
- Stable
- Degrading
- Unstable

6.2 Property Exploration Validation

Five targeted scenarios were designed:

- Near Identical
- Same Conclusion / Different Reasoning
- Contradictory Statements
- Synonym Variation
- Domain Shift

6.3 Cross-Domain Drift Validation

Four independent domain-drift trajectories were constructed:

- Finance
- Artificial Intelligence
- Medical
- Legal

Each trajectory progressively transitioned from a coherent reference field toward increasingly distant conceptual regions.

7. Results

7.1 Controlled Semantic Divergence Validation

Scenario	SFCS
Highly Stable	0.768
Stable	0.606
Degrading	0.331
Unstable	0.041

A monotonic decline was observed across all validation levels:

Plain Text
0.768
↓
0.606
↓
0.331
↓
0.041

The results support H1 and demonstrate sensitivity to increasing semantic divergence.

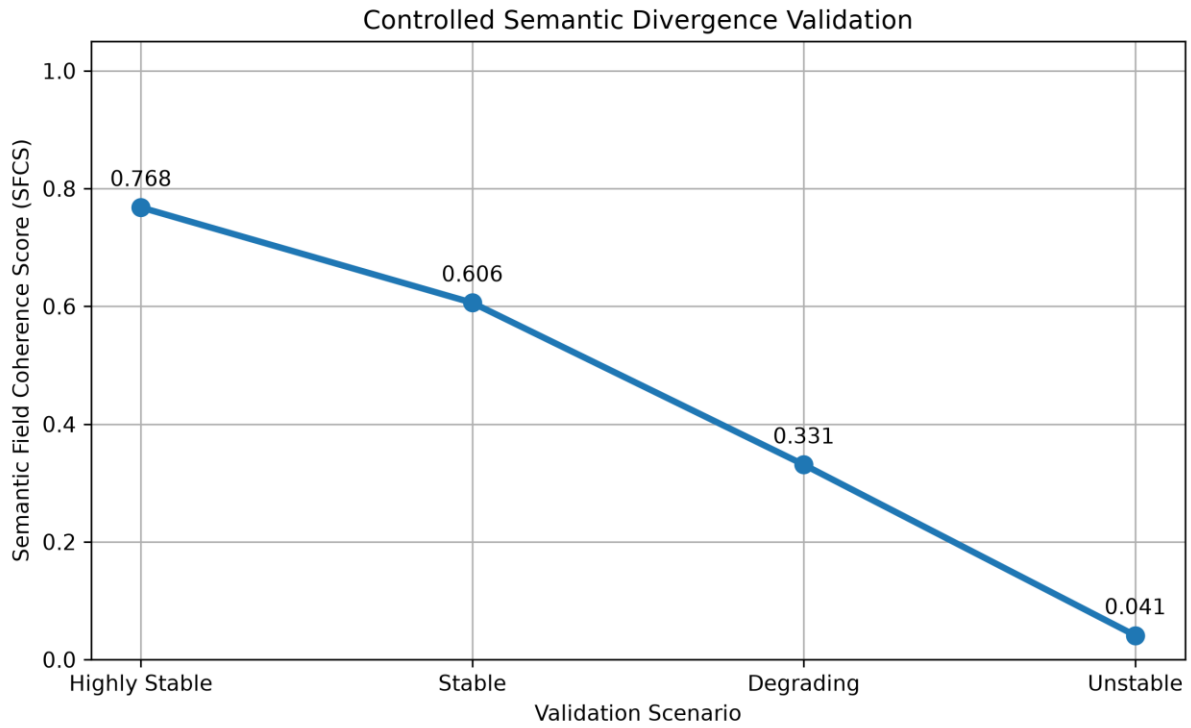


Figure 1. Controlled Semantic Divergence Validation.

SFCS decreases systematically as semantic divergence increases.

7.2 Property Exploration Validation

Scenario	SFCS
Near Identical	1.000
Same Conclusion Different Reasoning	0.782
Contradictory Statements	0.744
Synonyms	0.441
Domain Shift	0.291

Near-identical responses produced the expected upper-bound score of 1.000.

Outputs reaching the same conclusion through different reasoning paths maintained high coherence (0.782).

Contradictory statements produced a surprisingly high score (0.744), indicating preservation of semantic coherence despite logical opposition.

By contrast, domain-shift scenarios produced substantially lower coherence (0.291).

This contrast represents one of the most significant findings of the study.

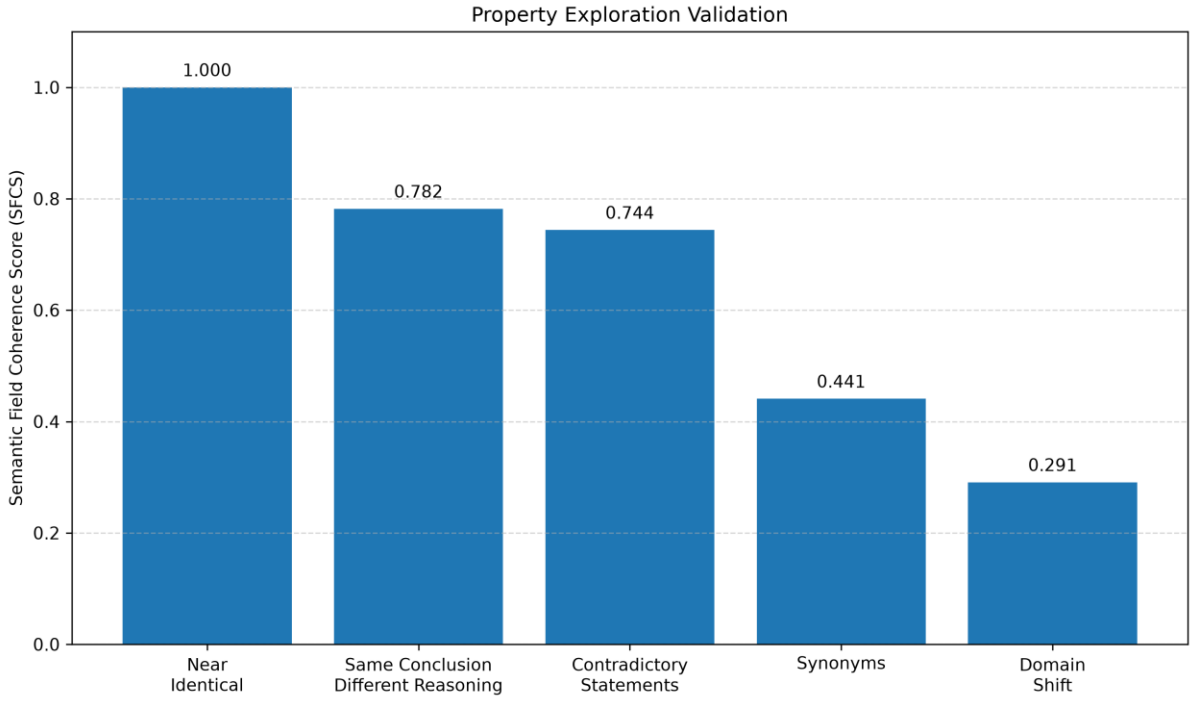


Figure 2. Property Exploration Validation.

Contradictory statements remain highly coherent while domain-shift scenarios produce substantial coherence degradation.

7.3 Cross-Domain Drift Validation

Four independent domain-drift trajectories were evaluated.

Finance

Reference:

Plain Text
0.572

Final:

Plain Text
-0.022

Total degradation:

Plain Text
0.594

Artificial Intelligence

Reference:

Plain Text
0.522

Final:

Plain Text
0.027

Total degradation:

Plain Text
0.495

Medical

Reference:

Plain Text
0.554

Final:

Plain Text
0.023

Total degradation:

Plain Text
0.531

Legal

Reference:

Plain Text
0.522

Final:

Plain Text
-0.023

Total degradation:

Plain Text
0.545

7.4 Cross-Domain Findings

All four domains exhibited substantial reductions in SFCS under progressive drift conditions.

Despite differences in domain content, a consistent pattern emerged:

Plain Text
High Coherence
↓
Drift
↓
Fragmentation
↓

Near-Zero Coherence

Normalized drift curves revealed comparable degradation dynamics across all four domains.

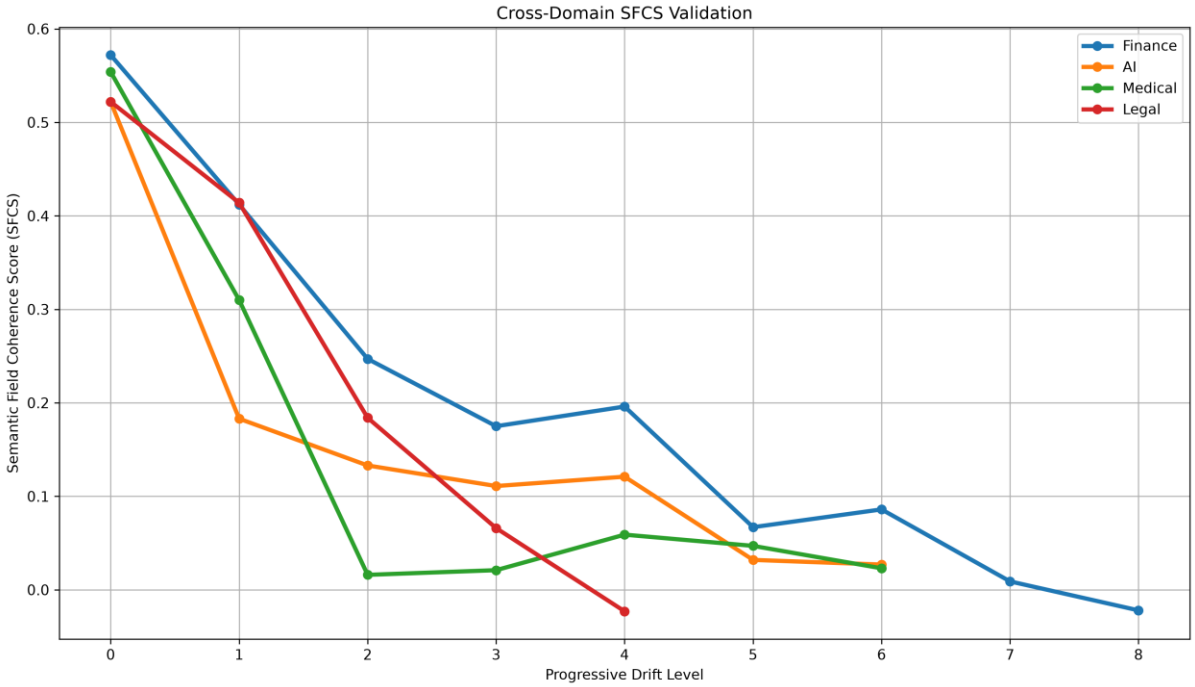


Figure 3. Cross-Domain SFCS Validation.
Finance, AI, medical, and legal drift trajectories all exhibit substantial coherence degradation under progressive semantic drift.

7.5 Summary of Findings

F1

SFCS exhibits systematic degradation under controlled semantic divergence.

F2

SFCS remains relatively high for contradictory statements that remain anchored within the same semantic field.

F3

SFCS decreases substantially under domain-shift conditions.

F4

Cross-domain experiments demonstrate comparable degradation trajectories across finance, AI, medical, and legal domains.

F5

SFCS appears sensitive to semantic field coherence rather than logical agreement.

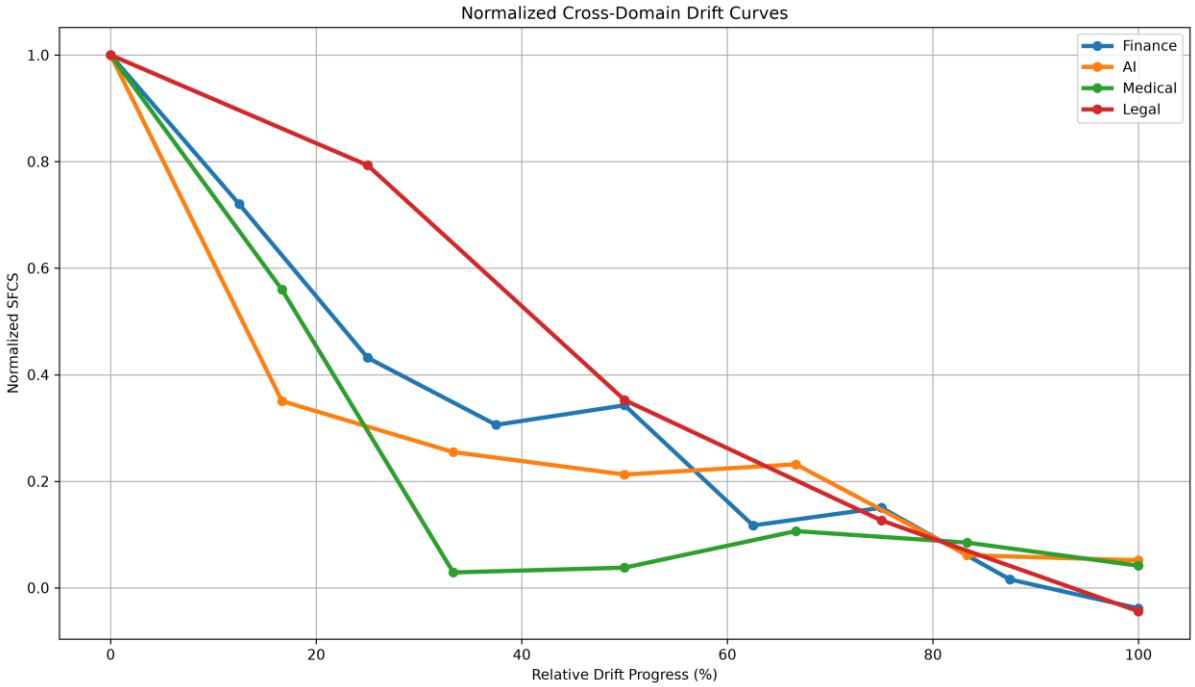


Figure 4. Normalized Cross-Domain Drift Curves.
Relative drift behavior remains comparable across multiple semantic domains despite differences in domain structure.

8. Discussion

8.1 Distinguishing Semantic Drift from Logical Disagreement

One of the most notable findings emerged from the contrast between the Contradictory Statements experiment and the Domain Shift experiment.

Contradictory statements maintained a high coherence value (0.744), while domain-shift scenarios produced substantially lower coherence (0.291).

For example:

Plain Text
Overfitting is harmful.

Overfitting is beneficial.

remain anchored within the same conceptual region despite expressing opposing claims.

These findings suggest that SFCS does not primarily measure logical agreement but rather the structural coherence of a shared semantic field.

This distinction may be particularly valuable in operational monitoring scenarios where semantic drift is of greater concern than local disagreement.

8.2 Semantic Field Degradation as an Early-Warning Signal

Across all drift trajectories, coherence degradation emerged gradually rather than as an abrupt failure event.

The observed trajectories suggest the following pattern:

Plain Text
Coherence
↓
Degradation
↓
Fragmentation
↓
Domain Drift
↓
Semantic Collapse

This observation raises the possibility that semantic instability may become observable before complete semantic collapse occurs.

If validated in real-world operational environments, semantic field degradation could potentially function as an early-warning signal for emerging domain drift and hallucination behavior.

9. Operational AI Assurance Applications

9.1 Domain Drift Monitoring

SFCS provides a direct signal for monitoring semantic field degradation and domain drift.

9.2 Long-Running Agent Supervision

Long-duration agent workflows may benefit from continuous semantic coherence monitoring.

9.3 Semantic Anomaly Detection

Sudden reductions in coherence may indicate abnormal semantic behavior or emerging instability.

9.4 Audit and Monitoring Dashboards

SFCS can be integrated into operational assurance dashboards, providing a lightweight and interpretable observability signal.

10. Limitations

10.1 Internal Validity & Embedding Model Sensitivity

All experiments were conducted using a single embedding architecture. Because SFCS relies directly on embedding-space geometry, baseline values and sensitivity thresholds may vary across models.

10.2 Construct Validity & Lexical Sensitivity

The Synonym Variation experiment yielded a lower coherence score than expected (0.441), indicating sensitivity to lexical shifts within the embedding space.

Future calibration may be required to distinguish stylistic variation from genuine semantic drift.

10.3 External Validity & Benchmark Scope

The validation scenarios were manually constructed controlled benchmarks.

While this approach was essential for isolating metric responsiveness, real-world deployment environments introduce substantially greater complexity and noise.

10.4 Alignment with Human Perception

Formal human-coherence alignment studies have not yet been conducted.

The relationship between SFCS values and human expert judgments remains an important area for future validation.

11. Future Research Directions

The present study represents an initial validation of SFCS as an observability metric for semantic field coherence. While the results are encouraging, several aspects require further investigation.

11.1 Evaluation Across Additional Embedding Models

The current experiments were conducted using a single embedding architecture. Future studies may investigate whether similar behavioral patterns emerge across alternative embedding models and vector representations. Such evaluations would help determine the robustness of SFCS across different semantic spaces.

11.2 Human Evaluation Studies

An important next step is the comparison of SFCS outputs with human judgments of semantic coherence and domain drift. Such studies may provide additional evidence regarding the relationship between measured coherence and human perception of semantic stability.

11.3 Real-World Validation

The experiments presented in this paper were intentionally designed as controlled validation scenarios. Future research may explore the behavior of SFCS in more complex environments, including longer interactions, operational AI systems, and real-world domain-shift conditions.

11.4 Operational Monitoring Applications

The observed sensitivity of SFCS to semantic degradation suggests potential applications in monitoring, auditing, and AI assurance workflows. Further investigation is needed to determine how SFCS behaves within practical observability pipelines and long-running AI systems.

11.5 Semantic Field Degradation and Hallucination Emergence

One particularly interesting question concerns the relationship between semantic field degradation and hallucination behavior. The results presented in this study suggest that semantic instability may emerge gradually rather than as an abrupt failure event. Future research may investigate whether degradation of semantic field coherence can function as an early-warning signal for emerging hallucinations or other forms of semantic failure.

12. Reproducibility and Research Artifacts

The experimental artifacts developed for this study include the SFCS Validation Workbench, domain-drift evaluation suites, validation datasets, result exports, and cross-domain validation scripts.

To support reproducibility and independent verification, the following artifacts are made available:

- SFCS Validation Workbench
- Finance Domain Drift Suite
- AI Domain Drift Suite
- Medical Domain Drift Suite
- Legal Domain Drift Suite
- Cross-Domain Validation Dashboard
- Validation datasets (CSV)
- Figure generation scripts
- Experimental result exports

The repository also contains the data used to generate the figures reported in this paper, enabling independent reproduction of the reported experimental results.

The complete implementation and experimental artifacts are available at:
https://github.com/<repo>
Repository contents include source code, validation datasets, experiment configurations, and figure-generation scripts.

13. Conclusion

This paper introduced the Semantic Field Coherence Score (SFCS), an observability metric designed to quantify semantic field coherence across AI outputs.

Controlled divergence experiments, contradiction analysis, domain-shift testing, and cross-domain validation demonstrated that SFCS responds systematically to semantic degradation while remaining relatively insensitive to logical disagreement within a shared semantic field.

Across finance, artificial intelligence, medical, and legal domains, SFCS consistently tracked semantic field degradation and domain drift.

These findings suggest that semantic field coherence can be transformed into a measurable engineering signal and monitored operationally. More broadly, the results indicate that semantic field degradation may become observable before complete semantic collapse occurs, opening new opportunities for AI observability, domain drift monitoring, operational assurance, and risk management in increasingly autonomous AI systems. The cross-domain validation results suggest that semantic field coherence may represent a domain-independent observability signal, although further evaluation across additional embedding architectures and operational environments is required.

The present study should be viewed as an initial validation of SFCS rather than a complete theory of semantic field dynamics.

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